BT4222 Group 13 — Dataset & Source-Code Listing

GitHub repository: https://github.com/steffanylars/BT4222_finalProj

1. Dataset

- **JDSearch (primary).** Liu et al., SIGIR 2023. 173,831 users, 12.8M products, 26.7M interactions. Source & download (JD Cloud): https://github.com/rucliujn/JDsearch · paper https://doi.org/10.1145/3539618.3591900
- **Files.** user_behavior_data.txt (919 MB; query, candidate_wid_list, candidate_label_list, history_qry/wid/type/time lists) and product_meta_data.txt (4.5 GB; wid, name, brand, 4 category levels, shop_id). Pre-processed to Parquet chunks under 'Interim Report/data/JDsearch/product_chunks/'.
- **Intermediate artifacts.** Final Project/artifacts/model/ (candidates, history, sessions, features, meta_subset .parquet) — git-ignored due to size; regenerate with src/step2 & step3.

2. Repository structure

- **Interim Report/** — interim deliverable, EDA notebooks, product_chunks, figures.
- **Final Project/src/** — modular pipeline step1–step6 (.py), make_report/make_slides/make_listing.
- **Final Project/report/** — BT4222_Group13_Final_Report.pdf (+ .docx) and this listing.
- **Final Project/slides/** — BT4222_Group13_Final_Presentation.pptx.
- **Final Project/figures/** — all figures (temporal, retention, model comparison, importance, segments).
- **Final Project/artifacts/** — result tables (JSON/CSV: temporal_summary, retention_cutoffs, results_*, segment_metrics, rank_by_type, ablation, lgb_feature_importance) + trained models (lgb_model.txt, dhar.pt).
- **Final Project/README.md & RESULTS.md** — run order, scaling knob, and preserved step outputs.

3. Source code (run in order)

- **step1_preprocess_temporal.py** — full-data temporal retention analysis (clicks 7d / carts 90d / orders 1yr)
- **step1b_retention_cutoffs.py** — exact 1-day-resolution cut-offs

- **step2_build_dataset.py [N_USERS]** — build sampled modelling dataset + metadata join
- **step3_features.py** — horizon-aware feature engineering
- **step4_baselines_lgbm.py** — Random/Popularity/Content/MF + LambdaMART, position-leak-safe eval
- **step5_neural.py** — Dual-Horizon Attention Ranker (ours) + NCF baseline (PyTorch/MPS)
- **step6_diagnostics.py** — segment analysis, rank-by-type, temporal-feature ablation